Experiment 15

Aim

To study the different career pathways available in low-code development and understand how low-code platforms help individuals build software applications, automate business processes, improve productivity, and develop technical skills for employment and entrepreneurship. The study also aims to identify the skills required to pursue a career in low-code development.

Introduction

Low-code development is an approach to software development that uses visual tools, drag-and-drop components, predefined templates, and minimal programming. Platforms such as Zoho Creator, Microsoft Power Apps, and OutSystems enable developers to build applications quickly and automate business processes. Low-code development is becoming useful in organizations that want to reduce development time, improve productivity, and automate repetitive tasks. It creates career opportunities for both technical and non-technical professionals.

Career Pathways

1. Low-Code Application Developer

A low-code application developer creates business applications using visual development platforms. Responsibilities include designing forms, creating reports, implementing workflows, and testing application functionality. For example, a developer can build a library book issue management application in Zoho Creator to register members, receive book issue requests, track returns, and manage fines.

2. Business Process Automation Specialist

These professionals identify repetitive business activities and automate them using workflows, scheduled tasks, notifications, and approval processes. In a library application, this role could involve automating issue confirmations, due-date reminders, overdue notices, and fine notifications.

3. Low-Code Solution Architect

A solution architect plans the structure of an application, selects suitable technologies, designs integrations, and ensures that the system meets business requirements. For a library, the architect might plan how member requests, the book catalog, fines, payments, and analytics should work together.

4. Business Analyst

Business analysts understand business requirements and translate them into application features. They communicate with users, document workflows, and help developers design appropriate solutions. For example, they may identify the information needed to register members and determine how an issue request should move from request to approval, issue, and return.

5. Integration and Pro-Code Developer

Low-code applications sometimes require custom programming or external integrations. Developers with skills in JavaScript, Node.js, Python, APIs, and databases can extend the capabilities of low-code platforms. For example, a library application could connect to an external catalog system or use a custom API to calculate overdue fines.

6. Low-Code Consultant and Administrator

Consultants help organizations select and implement low-code platforms. Administrators manage users, permissions, application configuration, monitoring, and maintenance. They also help maintain data security and ensure that applications continue to operate reliably.

Skills Required for a Career in Low-Code Development

- **Platform knowledge:** Understanding tools such as Zoho Creator and their development features.
- **Database fundamentals:** Learning how to store, retrieve, and manage application data.
- **Basic programming:** Understanding logic, conditions, loops, and scripting languages such as Deluge.
- **Problem-solving:** Identifying business problems and creating suitable application solutions.
- **Testing and debugging:** Finding errors and ensuring that applications work as expected.
- **Communication skills:** Understanding user requirements and explaining technical solutions.

Future of Low-Code Development

1. **Integration with Artificial Intelligence:** Low-code platforms are increasingly incorporating AI-assisted development, data analysis, and intelligent automation. These capabilities can help developers generate application components, analyze business data, and reduce manual work.
2. **Growth of Business Automation:** Organizations can use low-code tools to automate approvals, notifications, customer management, inventory processes, and reporting. This is expected to remain an important application area.
3. **Combination of Low-Code and Pro-Code:** Future applications will often combine visual development with custom programming. Low-code platforms can handle standard forms and workflows, while traditional code can support specialized calculations, advanced integrations, and complex processing.
4. **Greater Focus on Security and Governance:** As organizations develop more applications, managing user permissions, protecting sensitive data, monitoring changes, and maintaining compliance will become increasingly important.
5. **Cloud-Based Application Development:** Cloud platforms make applications accessible across locations and devices. Low-code development can help organizations deploy and maintain business applications without managing every infrastructure component themselves.
6. **Continued Need for Technical Skills:** Low-code does not eliminate the need for programming knowledge. Developers must still understand application logic, data structures, testing, integration, and security to build reliable systems.

Career Charts

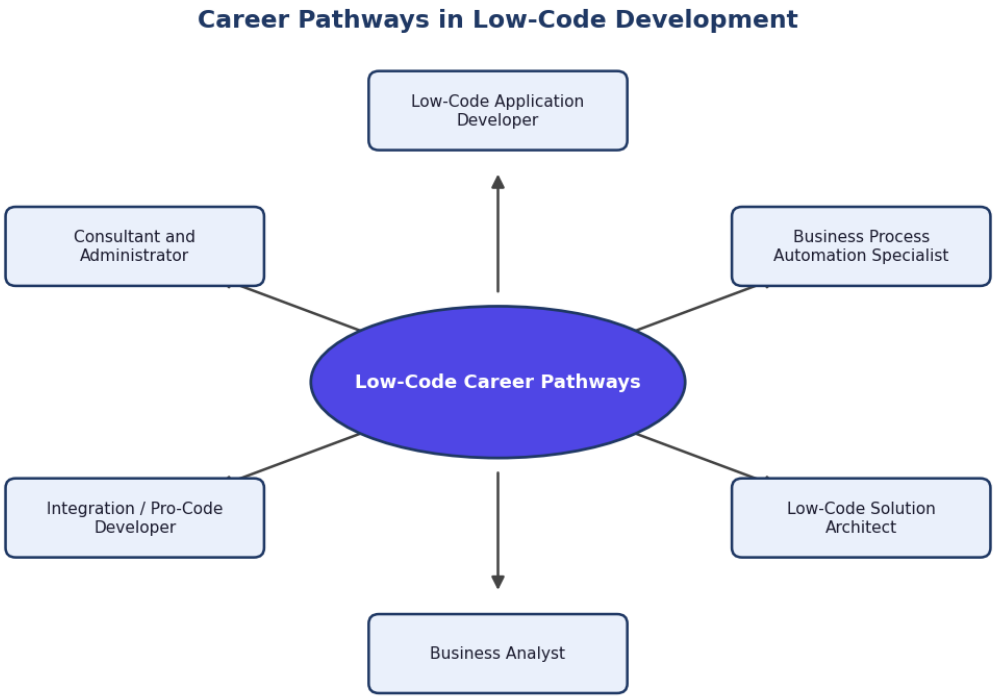


Figure 15-1: Career pathways in low-code development

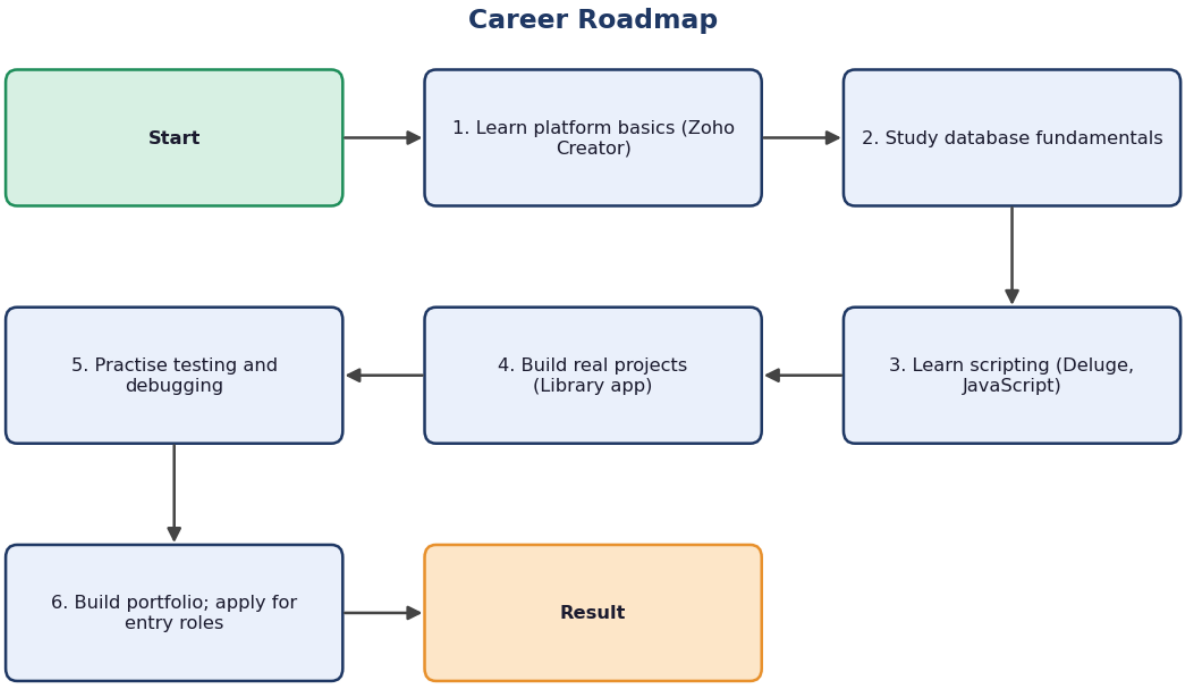


Figure 15-2: Career roadmap

Conclusion

Low-code development offers career opportunities in application development, workflow automation, software testing, consulting, and freelancing. These careers allow individuals to develop useful business applications with less manual coding and reduced development time.

Low-code platforms such as Zoho Creator are suitable for beginners because they provide visual development tools and built-in features for forms, reports, workflows, and dashboards. By gaining practical experience, learning database concepts, developing basic scripting skills, and completing real-world projects such as the Library Book Issue Management application, students can build a strong portfolio and prepare for entry-level IT jobs. Combining low-code knowledge with programming, API integration, and AI tools can further expand career opportunities.